

Multivariable calculus

1. Vectors, matrices

Definition 1.1 (Vector length)

The **length** or **magnitude** of a vector $\begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix}$ in $\mathbb{R}^n$ is

$$\left\| \begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix} \right\| = \left\| \begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix} \right\| := \sqrt{v_1^2 + v_2^2 + \dots}$$

Definition 1.2 (Displacement vector)

The vector from the end of $\vec{A}$ to the end of $\vec{B}$ when their starts are in the same location.

$$\overrightarrow{AB} = \vec{B} - \vec{A}$$

1.1. Dot product

Definition 1.1.1 (Dot product)

Where $\vec{a}, \vec{b} \in \mathbb{R}^n$ and θ is the angle between $\vec{a}$ and $\vec{b}$:

$$\vec{a} \cdot \vec{b} = \sum a_i b_i = |\vec{a}| |\vec{b}| \cos \theta$$

Also called **scalar product** or **inner product**.

Theorem 1.1.2 (Properties of dot product)

For $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^n$ and $k \in \mathbb{R}$:

- $\vec{a} \cdot \vec{a} = |\vec{a}|^2$
- $\vec{a} \cdot \vec{a} = 0$ iff $\vec{a} = 0$
- Commutativity: $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
- Distributivity: $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$
- Distributivity: $(k\vec{a}) \cdot \vec{b} = k(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (k\vec{b})$
- $\vec{a} \cdot \vec{b} = 0$ iff $\vec{a} \perp \vec{b}$, $\vec{a} = 0$, or $\vec{b} = 0$.

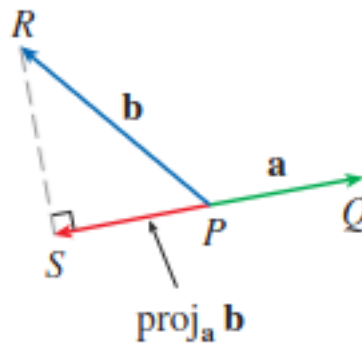
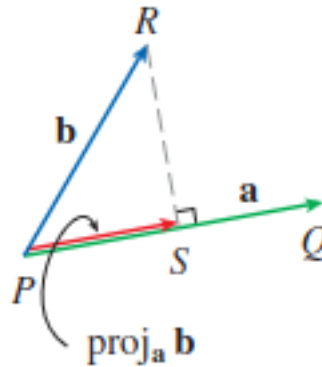
Definition 1.1.3 (Projection)

The **scalar projection of $\vec{b}$ onto $\vec{a}$** or **component of $\vec{b}$ along $\vec{a}$** , is defined as (where $\theta \in [0, \pi]$ is the angle between $\vec{a}$ and $\vec{b}$):

$$\text{comp}_{\vec{a}} \vec{b} := \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} = |\vec{b}| \cos(\theta)$$

The **vector projection of $\vec{b}$ onto $\vec{a}$** , is defined as:

$$\text{proj}_{\vec{a}} \vec{b} := \text{comp}_{\vec{a}} \vec{b} \frac{\vec{a}}{|\vec{a}|} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \vec{a}$$



1.2. Cross product

See Stewart 12.4, NMD Lecture 4

Definition 1.2.1 (Cross product)

For $\vec{a}, \vec{b} \in \mathbb{R}^3$, the unique vector $\vec{a} \times \vec{b}$ satisfying

- $|\vec{a} \times \vec{b}|$ is the area of the parallelogram spanned by $\vec{a}$ and $\vec{b}$
- $\vec{a} \times \vec{b} = 0$ iff $\vec{a} \parallel \vec{b}$, $\vec{a} = 0$, or $\vec{b} = 0$.
- $\vec{a} \times \vec{b}$ is orthogonal to $\vec{a}$ and $\vec{b}$.
- $(\vec{a}, \vec{b}, \vec{a} \times \vec{b})$ is right-handed (if the coordinate system is right-handed)

Theorem 1.2.2 (Properties of cross product)

For $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^3$ and $k \in \mathbb{R}$:

- $\vec{a} \times \vec{b} = (-\vec{b}) \times \vec{a}$
- $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$
- $(\vec{a} + \vec{b}) \times \vec{c} = \vec{a} \times \vec{c} + \vec{b} \times \vec{c}$
- $k(\vec{a} \times \vec{b}) = (k\vec{a}) \times \vec{b} = \vec{a} \times (k\vec{b})$
- $\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c}$
- $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

Theorem 1.2.3 (Calculation of cross product)

Where $\vec{a}, \vec{b} \in \mathbb{R}^3$ and θ is the angle between $\vec{a}$ and $\vec{b}$:

$$\begin{aligned} \vec{a} \times \vec{b} &= \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \\ &= \hat{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \hat{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \hat{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \\ | \vec{a} \times \vec{b} | &= |\vec{a}| |\vec{b}| \sin \theta \end{aligned}$$

1.2.1. Parallelograms and parallelepipeds

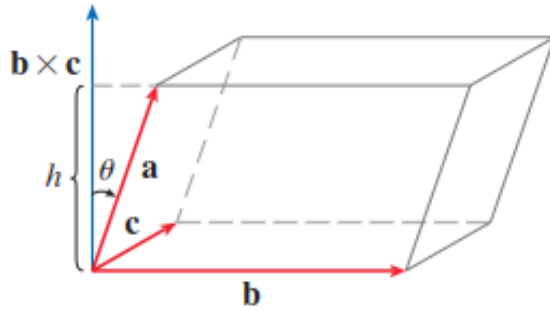
Theorem 1.2.4 (Parallelogram spanned by vectors in $\mathbb{R}^2$)

The area of the parallelogram spanned by the vectors $\vec{v}, \vec{w} \in \mathbb{R}^2$ is given by

$$\begin{vmatrix} v_1 & v_2 \\ w_1 & w_2 \end{vmatrix} = v_1 w_2 - v_2 w_1.$$

The sign of the result is positive if $\vec{w}$ is counterclockwise from $\vec{v}$ and negative if clockwise.

Theorem 1.2.5 (Volume of parallelepiped)



Consider the parallelepiped determined by $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^3$, shown above. The area of its base is $|\vec{b} \times \vec{c}|$. Let θ be the angle between $\vec{a}$ and $\vec{b} \times \vec{c}$; then, its height is $|\vec{a}||\cos \theta|$. Thus, the volume of the parallelepiped is

$$V = Ah = |\vec{b} \times \vec{c}||\vec{a}||\cos \theta| = |\vec{a} \times (\vec{b} \times \vec{c})|.$$

1.3. Matrices

Definition 1.3.1 (Matrix multiplication)

The matrix multiplication of the $m \times n$ matrix A and the $n \times p$ matrix B is made by dot-producting the rows of the first by the columns of the second:

$$[AB]_{ij} = [A_{i*} \cdot B_{*j}] = \left[\sum_{k=1}^n A_{ik} B_{kj} \right]$$

Definition 1.3.2 (Positive definite)

Let A be a matrix. Then A is positive definite iff any of the following equivalent conditions are true:

- For all $v \in \mathbb{R}^n \setminus \{0\}$, $v^T A v > 0$
- All leading principal minors of A are positive
- All eigenvalues of A are positive
- All pivots of A are positive

Definition 1.3.3 (Negative definite)

Let A be a matrix. Then A is negative definite iff any of the following equivalent conditions are true:

- For all $v \in \mathbb{R}^n \setminus \{0\}$, $v^T A v < 0$
- The k th-order leading principal minor is negative if k is odd and positive if k is even
- All eigenvalues of A are negative
- All pivots of A are negative
- $-A$ is positive definite

Positive semidefinite and **negative semidefinite** are the same except that the determinant/eigenvalue/pivot/ $v^T A v$ could also be 0.

Definition 1.3.4 (Principal minor)

The determinant of a submatrix of a matrix.

Definition 1.3.5 (Leading principal minor)

The k th-order leading principal minor is the determinant of the top left submatrix of a matrix, where the 1st-order leading principal minor is the determinant of the 1x1 matrix at its top left corner, the 2nd-order is the determinant of the 2x2 matrix at its top left corner, etc.

1.4. Useful properties

Theorem 1.4.1 (Cauchy-Schwarz inequality)

For all $\vec{x}, \vec{y} \in \mathbb{R}^n$,

$$|\vec{x} \cdot \vec{y}| \leq |\vec{x}||\vec{y}|.$$

Theorem 1.4.2 (Triangle inequality)

For all $\vec{x}, \vec{y} \in \mathbb{R}^n$,

$$|\vec{x} + \vec{y}| \leq |\vec{x}| + |\vec{y}|.$$

2. Lines and planes

2.1. Lines

See Stewart [12.5]

The most useful notation for a line is in parametric form:

Definition 2.1.1 (Parametric form of a line)

Where $\vec{r}_0 \in \mathbb{R}^3$ is a point on the line, and $t \in \mathbb{R}^3$:

$$\vec{r}(t) = \vec{r}_0 + \vec{v}t$$

$\vec{v}$ is called the **direction vector**. Then, every value of t gives a point $\vec{r}(t)$ on the line.

Theorem 2.1.2

Two lines are parallel iff their direction vectors are scalar multiples of each other.

Definition 2.1.3 (Skew lines)

Two lines are skew iff they do not intersect but are not parallel, i.e. they lie in different parallel planes.

2.2. Planes

See Stewart [12.5]

Theorem 2.2.1 (Vector equation for a plane in $\mathbb{R}^3$)

If $\vec{n} \in \mathbb{R}^3$ is the **normal vector** to the plane (vector perpendicular to the plane), and $\vec{P} \in \mathbb{R}^3$ is a point on the plane, then any other point $\vec{r} \in \mathbb{R}^3$ satisfies:

$$\vec{n} \cdot (\vec{r} - \vec{P}) = 0$$

or equivalently:

$$\vec{n} \cdot \vec{r} = \vec{n} \cdot \vec{P}$$

We can also write it as a scalar equation:

$$n_x(x - P_x) + n_y(y - P_y) + n_z(z - P_z) = 0$$

Theorem 2.2.2

Two planes are parallel iff their normal vectors are parallel.

Theorem 2.2.3 (Equation of plane containing three points)

If $A, B, C \in \mathbb{R}^3$ are points on our plane, then we can find the normal vector by performing $n = \overrightarrow{AB} \times \overrightarrow{AC} = (B - A) \times (C - A)$ (since $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are vectors on the plane).

Theorem 2.2.4 (Parametric equation for a plane in $\mathbb{R}^3$)

If $a, b \in \mathbb{R}^3$ are nonparallel nonzero vectors on the plane, and $P \in \mathbb{R}^3$ is a point on the plane, then the parametric equation for the plane is:

$$\vec{x}(s, t) = \vec{P} + s\vec{a} + t\vec{b}$$

2.3. Distance

See Stewart [12.5]

Procedure 2.3.1 (Distance between point and line)

Let P be the point, and $A + Lt$ be the line. Then the distance is

$$|\overrightarrow{AP} - \text{proj}_L \overrightarrow{AP}| = \vec{P}$$

Procedure 2.3.2 (Distance between parallel planes)

Let Π_1 and Π_2 be the two planes.

If n is normal to both planes, and $P_1 \in \Pi_1$ and $P_2 \in \Pi_2$, then the answer is

$$|\text{proj}_n \overrightarrow{P_1 P_2}|$$

3. Functions and quadric surfaces

4. Limits

Definition 4.1 (Limit)

If $f : X \rightarrow \mathbb{R}$, where $X \subseteq \mathbb{R}^n$, then $\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) = L$ iff for every $\varepsilon > 0$ there exists $\delta > 0$ s.t. $\vec{x} \in X$ and $0 < |\vec{x} - \vec{a}| < \delta$ implies $|f(\vec{x}) - L| < \varepsilon$.

Warning: The limit might be different along different paths to $\vec{a}$! If different paths result in different limits, then there is no limit.

Definition 4.2 (Continuity)

If $f : X \rightarrow \mathbb{R}$, where $X \subseteq \mathbb{R}^n$, then f is **continuous** at $\vec{a}$ iff

$$\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) = f(\vec{a}).$$

f is **continuous** on X iff f is continuous at every $\vec{a} \in X$.

Theorem 4.3 (Limit laws)

$$\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) + g(\vec{x}) = \lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) + \lim_{\vec{x} \rightarrow \vec{a}} g(\vec{x})$$

$$\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x})g(\vec{x}) = \left(\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) \right) \left(\lim_{\vec{x} \rightarrow \vec{a}} g(\vec{x}) \right)$$

$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{f(\vec{x})}{g(\vec{x})} = \frac{\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x})}{\lim_{\vec{x} \rightarrow \vec{a}} g(\vec{x})}$$

Theorem 4.4 (Squeeze theorem)

If

- $g(\vec{x}) \leq f(\vec{x}) \leq h(\vec{x})$ for all $\vec{x}$ in an open ball containing $\vec{a}$, except possibly at $\vec{a}$ itself, and
- $\lim_{\vec{x} \rightarrow \vec{a}} g(\vec{x}) = \lim_{\vec{x} \rightarrow \vec{a}} h(\vec{x}) = L$,

then $\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) = L$.

5. Differentiation

5.1. Partial derivatives

5.2. Differentiability

Definition 5.2.1 (Linear approximation ($\mathbb{R}^n \rightarrow \mathbb{R}$))

The **linear approximation** or **tangent plane** ($\mathbb{R}^3$) or **hyperplane** to the graph of a function f at the point $\vec{a}$ is expressed by

$$L(\vec{x}) = f(\vec{a}) + \nabla f(\vec{a}) \cdot (\vec{x} - \vec{a})$$

In $\mathbb{R}^3$, this is equivalent to the plane

$$\begin{aligned} z = L(x, y) &= f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b) \\ &= f(\vec{a}) + f_x(\vec{a})\Delta x + f_y(\vec{a})\Delta y. \end{aligned}$$

Definition 5.2.2 (Linear approximation ($\mathbb{R}^n \rightarrow \mathbb{R}^m$))

The **linear approximation** to a vector-valued function f at the point $\vec{a}$ is expressed by

$$L(\vec{x}) = f(\vec{a}) + Jf(\vec{a})(\vec{x} - \vec{a})$$

Definition 5.2.3 (Differentiability)

Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$, where X is an open subset of $\mathbb{R}^n$, and let $\vec{a} \in X$. f is differentiable at a iff all of its partial derivatives exist and

$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{f(\vec{x}) - L(\vec{x})}{|\vec{x} - \vec{a}|} = 0$$

where $L(\vec{x})$ is the linear approximation to f at $\vec{a}$.

If $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, then we can instead say f is differentiable iff

$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{\Delta z - (f_x(\vec{a})\Delta x + f_y(\vec{a})\Delta y)}{|\vec{x} - \vec{a}|} = 0.$$

Theorem 5.2.4

Differentiability implies continuity.

Theorem 5.2.5 (Differentiability shortcut)

Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a vector-valued function. If all partial derivatives $\frac{\partial f_i}{\partial x_j}$ exist and are continuous in an open ball around $\vec{a}$ in X , then F is differentiable at $\vec{a}$.

5.3. Chain rule

See Colley [2.5], Trimm [3.8, 6.5, DiffEq-1.0], Brummet [12, MVCWUP:Feb24(58-61)]

See NMD [10], Stewart [14.5]

Theorem 5.3.1 (Multivariable chain rule)

Suppose $X \subseteq \mathbb{R}^m$ and $T \subseteq \mathbb{R}^n$ are open and $f : X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^p$ and $r : T \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ are defined so that $r(T) \subseteq X$. We wish to find the derivative of

$$f \circ r : T \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m \rightarrow \mathbb{R}^p.$$

If r is differentiable at $t_0 \in T$ and f is differentiable at $x_0 = r(t_0)$, then the composite $f \circ r$ is differentiable at t_0 , and we have

$$J(f \circ r)(t_0) = Jf(x_0)Jr(t_0).$$

In $\mathbb{R}^3$, if $n = 1$, $p = 1$, and $m = 3$ (i.e. $f \circ r : T \subseteq \mathbb{R} \rightarrow \mathbb{R}^3 \rightarrow \mathbb{R}$), define $(x, y, z) := (x_1, x_2, x_3)$. Then,

$$\frac{d(f \circ r)}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}.$$

5.4. Gradient and Jacobian

Definition 5.4.1 (Gradient)

$$\nabla f(\vec{x}) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{bmatrix}$$

Theorem 5.4.2

$\nabla f(\vec{a})$ is orthogonal to the level set of f at $\vec{a}$.

Definition 5.4.3 (Jacobian)

If $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a vector-valued function, then the Jacobian is

$$Jf \left(\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \right) = \begin{bmatrix} \nabla f_1 \\ \nabla f_2 \\ \vdots \\ \nabla f_m \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \dots & \frac{\partial f}{\partial x_n} \end{bmatrix}$$
$$= \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}.$$

The Jacobian Jf is also called Df or ∇f .

5.5. Directional derivative

See Colley [2.6], Trimm [3.7], Brummet [14]

See NMD [11]

See <https://youtu.be/7JEWlfFoJJQ>

Definition 5.5.1 (Directional derivative)

Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, where X is an open subset of $\mathbb{R}^n$, and let $\vec{a} \in X$. If $\vec{v}$ is any unit vector in X , then the directional derivative of f at $\vec{a}$ in the direction of $\vec{v}$ is

$$D_{\vec{v}}f(\vec{a}) = \lim_{h \rightarrow 0} \frac{f(\vec{a} + h\vec{v}) - f(\vec{a})}{h}$$

We may wish to normalize the directional derivative:

$$D_{\vec{v}}f(\vec{a}) = \lim_{h \rightarrow 0} \frac{f(\vec{a} + h\vec{v}) - f(\vec{a})}{h|\vec{v}|}$$

Theorem 5.5.2

If f is differentiable at $\vec{a}$, then

$$D_{\vec{v}}f(\vec{a}) = \nabla f(\vec{a}) \cdot \vec{v}$$

We may wish to normalize the directional derivative:

$$D_{\vec{v}}f(\vec{a}) = \nabla f(\vec{a}) \cdot \frac{\vec{v}}{|\vec{v}|}$$

Theorem 5.5.3

The gradient points in the direction of steepest ascent, and its magnitude is the derivative in that direction:

$$|\nabla f(\vec{a})| = D_{\widehat{\nabla f(\vec{a})}}f(\vec{a}) = \max\{D_{\vec{v}}f(\vec{a}) : \vec{v} \in \mathbb{R}^n\}$$

where $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$.

Theorem 5.5.4

Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, and let $(\vec{a}, k) \in \mathbb{R}^{n+1}$. Then $\nabla f(\vec{a})$ is orthogonal to the level curve at height k .

Thus, the tangent plane to the surface described by $f(\vec{x}) = k$ at point $\vec{a}$ is the plane with normal vector $\nabla f(\vec{a})$.

Example. The electric potential V 's negative gradient is called $\vec{E}$:

$$\vec{E} = -\nabla V.$$

This is because in electrostatics, opposites attract, so positive charges want to follow the steepest descent to an area with low electric potential. ∇V points to the steepest ascent, so $-\nabla V$ points to the steepest descent.

Theorem 5.5.5

The second directional derivative of f in the direction of $\vec{v}$ is given by

$$\vec{v}^T (\mathbf{H}f) \vec{v}.$$

6. Extrema

See Colley [4.1], Brummet [15, MVCWUP:69-80(Mar 4-6)]

See NMD [12, 13], Stewart [14.7]

See <https://youtu.be/7JEWlFfoJJQ>

6.1. Local extrema

Definition 6.1.1 (Critical point of f)

A point $\vec{c}$ in the domain of f where all of the partial derivatives of f at $\vec{c}$ equal 0 or do not exist, or equivalently $\nabla f(\vec{c}) = 0$ or does not exist.

Definition 6.1.2 (Saddle point)

A critical point that is not a max or min.

Theorem 6.1.3 (by Fermat's Theorem)

If $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ has a local maximum or minimum at $\vec{a}$ and the first order partial derivatives exist, then $\vec{a}$ is a critical point.

Definition 6.1.4 (Hessian matrix)

The Hessian matrix $\mathbf{H}f$ of a function $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is the matrix of second-order partials

$$[\mathbf{H}f_{ij}] = \left[\frac{\partial^2 f}{\partial x_i \partial x_j} \right]$$

If $X \subseteq \mathbb{R}^2$, then

$$\mathbf{H}f = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

The Hessian is the transpose of the Jacobian of the gradient:

$$\mathbf{H}f = (\mathbf{J}\nabla f)^T.$$

Theorem 6.1.5 (Second partial derivative test)

Let X be an open subset of $\mathbb{R}^n$ and $f : X \rightarrow \mathbb{R}$ whose 2nd-order and lower partials exist and are continuous on X (f is of class C^2). Let $\vec{a} \in X$ be a critical point of f . Then

- If the Hessian $\mathbf{H}f(\vec{a})$ is positive definite, then f has a local minimum at $\vec{a}$.
 - ▶ Easiest way to check is that the leading principal minors (determinant of the top left submatrices) are positive.
 - ▶ This is equivalent to the second derivative $\vec{v}^T (\mathbf{H}f) \vec{v}$ being positive.
- If the Hessian $\mathbf{H}f(\vec{a})$ is negative definite, then f has a local maximum at $\vec{a}$.
 - ▶ Easiest way to check is that the 1st leading principal minor is negative, 2nd is positive, 3rd is negative, etc.
 - ▶ This is equivalent to the second derivative $\vec{v}^T (\mathbf{H}f) \vec{v}$ being positive.
- If $\det \mathbf{H}f(\vec{a}) \neq 0$ but $\mathbf{H}f(\vec{a})$ is neither positive nor negative definite, then f has a saddle point at $\vec{a}$.
- If the Hessian has both positive and negative eigenvalues, then f has a saddle point at $\vec{a}$.
- Otherwise, the Hessian must be positive semidefinite or negative semidefinite, and the test is inconclusive.

If $X \subseteq \mathbb{R}^2$, let

$$D := f_{xx}(\vec{a})f_{yy}(\vec{a}) - (f_{xy}(\vec{a}))^2 = \det \mathbf{H}f(\vec{a})$$

Then

- If $D > 0$ and $f_{xx}(\vec{a}) > 0$, then f has a local minimum at $\vec{a}$.
- If $D > 0$ and $f_{xx}(\vec{a}) < 0$, then f has a local maximum at $\vec{a}$.
- If $D < 0$, then f has a saddle point at $\vec{a}$.
- If $D = 0$ the test is inconclusive.

(Note that if $f_{xx}(\vec{a}) = 0$ then $D \leq 0$.)

6.2. Taylor polynomials

See <https://youtu.be/V4iscOnH6W8>

The **first-order Taylor polynomial** is just the linear approximation

$$T_1(\vec{x}) = f(\vec{a}) + \nabla f(\vec{a}) \cdot (\vec{x} - \vec{a})$$

Definition 6.2.1 (Second-order Taylor polynomial)

The **second-order Taylor polynomial** for a function $f \in \mathbb{R}^n \rightarrow \mathbb{R}$ at point $\vec{a}$ evaluated at point $\vec{x}$, where $\vec{h} := \vec{x} - \vec{a}$, is:

$$\begin{aligned} T_2(\vec{x}) &= f(\vec{a}) + \sum_{i=1}^n f_{x_i}(\vec{a})h_i + \frac{1}{2} \sum_{i,j=1}^n f_{x_i x_j}(\vec{a})h_i h_j \\ &= f(\vec{a}) + \nabla f(\vec{a}) \cdot \vec{h} + \frac{1}{2} \vec{h}^T (\mathbf{H}f(\vec{a})) \vec{h} \end{aligned}$$

Higher-order Taylor polynomials are not very useful.

6.3. Absolute extrema

Theorem 6.3.1 (Extreme Value Theorem)

Let X be a closed and bounded subset of $\mathbb{R}^n$ and suppose $f : X \rightarrow \mathbb{R}^n$ is continuous. Then f attains an absolute maximum and an absolute minimum somewhere on X .

Theorem 6.3.2 (Method to find absolute minima and maxima with a constraint)

Let A be a closed bounded subset of $\mathbb{R}^n$, and C be the set of all critical points of f contained within A . Let $S := C \cup \partial A$, the union of C and the boundary of A . Then, the absolute maximum is $\max\{f(\vec{c}) : \vec{c} \in S\}$ and the absolute minimum is $\min\{f(\vec{c}) : \vec{c} \in S\}$.

6.4. Lagrange multiplier

See <https://www.youtube.com/watch?v=8mjcnxGMwFo>

Theorem 6.4.1

If $f(\vec{x}_0) = c$ is an extreme value (absolute max or min) of f on the set $\{\vec{x} : g(\vec{x}) = k\}$ and $\nabla g(\vec{x}_0) \neq 0$, then the following equivalent things are true:

- at $\vec{x}_0$, the level set $\{\vec{x} : f(\vec{x}) = c\}$ is tangent to $g(\vec{x}) = k$.
- $\nabla f(\vec{x}_0) = \lambda \nabla g(\vec{x}_0)$, where $\lambda \in \mathbb{R}$ is called the **Lagrange multiplier**.

Procedure 6.4.2

To find the extreme values of f on the set $\{\vec{x} : g(\vec{x}) = k\}$, solve the system of equations

$$\begin{aligned}\nabla f(\vec{x}) &= \lambda \nabla g(\vec{x}) \\ g(\vec{x}) &= k.\end{aligned}$$

7. Vector functions

See <https://youtu.be/40r56pX4mqA>, <https://youtu.be/80J5s0pic8M>

See Brummet [MVCWUP:Feb24(55-56)]

See NMD [15], Stewart [13.1-13.4]

Definition 7.1 (Path)

A **path** or **curve** in $\mathbb{R}^n$ is the image of a function $f : I \subseteq \mathbb{R} \rightarrow \mathbb{R}^n$, i.e. it is the set $f(I)$. If $I = [a, b]$, then the endpoints of the path are $f(a)$ and $f(b)$.

Definition 7.2 (Derivative of vector-valued function)

Let $\vec{f} : T \subseteq \mathbb{R} \rightarrow \mathbb{R}^m$. Then

$$\vec{f}'(t) := \begin{bmatrix} f_1'(t) \\ f_2'(t) \\ \vdots \\ f_m'(t) \end{bmatrix}$$

Theorem 7.3 (Differentiation rules)

Let $\vec{u}, \vec{v} : \mathbb{R} \rightarrow \mathbb{R}^n$, $f : \mathbb{R} \rightarrow \mathbb{R}$, $c \in \mathbb{R}$. Then,

$$\frac{d}{dt} \vec{u}(t) + \vec{v}(t) = \vec{u}'(t) + \vec{v}'(t)$$

$$\frac{d}{dt} c\vec{u}(t) = c\vec{u}'(t)$$

$$\frac{d}{dt} f(t)\vec{u}(t) = f'(t)\vec{u}(t) + f(t)\vec{u}'(t)$$

$$\frac{d}{dt} \vec{u}(t) \cdot \vec{v}(t) = \vec{u}'(t) \cdot \vec{v}(t) + \vec{u}(t) \cdot \vec{v}'(t)$$

$$\frac{d}{dt} \vec{u}(t) \times \vec{v}(t) = \vec{u}'(t) \times \vec{v}(t) + \vec{u}(t) \times \vec{v}'(t)$$

$$\frac{d}{dt} \vec{u}(f(t)) = f'(t)\vec{u}'(f(t))$$

Definition 7.4 (Tangent vector)

Given a path $\vec{r}(I)$ given by $\vec{r} : I \subseteq \mathbb{R} \rightarrow \mathbb{R}^3$, the tangent vector to said path at some point P is given by $\vec{r}'(t)$, provided that $\vec{r}'(t) \neq 0$.

Definition 7.5 (Limit of vector-valued function)

Let $\vec{f} : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$. Then

$$\lim \vec{f}(t) := \begin{bmatrix} \lim f_1(t) \\ \lim f_2(t) \\ \vdots \\ \lim f_m(t) \end{bmatrix}$$

Theorem 7.6

$\vec{f} : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous at $\vec{a}$ iff the components are continuous.

7.1. Arc length

Theorem 7.1.1 (Arc length)

The length of a path $\vec{r}([a, b])$ given by $\vec{r} : [a, b] \subseteq \mathbb{R} \rightarrow \mathbb{R}^3$ is

$$L = \int_a^b |\vec{r}'(t)| dt$$

Definition 7.1.2 (Arc length function)

Let $\vec{r}(I)$ be a path given by $\vec{r} : I \subseteq \mathbb{R} \rightarrow \mathbb{R}^3$. Then the **arc length function** from a , which takes in t and returns the arc length traversed since $f(a)$, is

$$s(t) := \int_a^t |\vec{r}'(u)| du$$

Procedure 7.1.3 (Reparameterizing in terms of arc length)

Let $\vec{r}(I)$ be a path given by $\vec{r} : I \subseteq \mathbb{R} \rightarrow \mathbb{R}^3$. Let $s(t)$ be the arc length function of $\vec{r}$ from a . Then, the reparameterization of $\vec{r}$ in terms of arc length from a is $\vec{r} \circ s^{-1}$.

7.2. Curvature and torsion

Definition 7.2.1 (Smooth)

A parameterization $\vec{r} : I \subseteq \mathbb{R} \rightarrow \mathbb{R}^3$ is **smooth** iff $\vec{r}'$ is continuous and $\vec{r}'(t) \neq 0$ for all $t \in I$.

A path is smooth iff it has a smooth parameterization.

Definition 7.2.2 (Unit tangent vector)

Let $\vec{r}(I)$ be a path given by $\vec{r} : I \subseteq \mathbb{R} \rightarrow \mathbb{R}^3$. Then, its **unit tangent vector** at $\vec{r}(t)$ is

$$\vec{T}(t) := \frac{\vec{r}'(t)}{|\vec{r}'(t)|}.$$

Definition 7.2.3 (Curvature)

Let $\vec{r}(I)$ be a path given by $\vec{r} : I \subseteq \mathbb{R} \rightarrow \mathbb{R}^3$. The **curvature** of $\vec{r}$ at t is the change in unit tangent vector per unit arc length:

$$\kappa(t) := \left| \frac{d\vec{T}}{ds} \right| = \frac{|\vec{T}'(t)|}{|\vec{r}'(t)|} = \frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^3}.$$

Definition 7.2.4 (Principal unit normal vector)

Let $\vec{r}(I)$ be a path given by $\vec{r} : I \subseteq \mathbb{R} \rightarrow \mathbb{R}^3$. Let t be a point s.t. $\kappa(t) \neq 0$. The **principal unit normal vector** (also called **unit normal**) of $\vec{r}$ at t is

$$\vec{N}(t) := \frac{\vec{T}'(t)}{|\vec{T}'(t)|}.$$

It is always perpendicular to $\vec{T}(t)$, and indicates the direction in which the curve is turning at each point.

Definition 7.2.5 (Binormal vector)

Let $\vec{r}(I)$ be a path given by $\vec{r} : I \subseteq \mathbb{R} \rightarrow \mathbb{R}^3$. Let t be a point s.t. $\kappa(t) \neq 0$. The **binormal vector** of $\vec{r}$ at t is

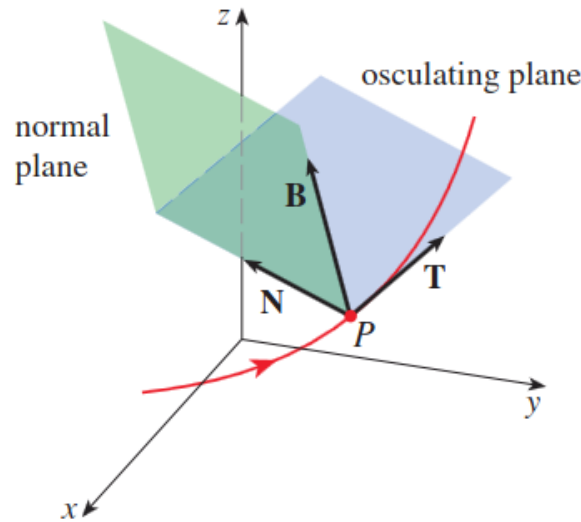
$$\vec{B}(t) = \vec{T}(t) \times \vec{N}(t).$$

The binormal vector is perpendicular to both $\vec{T}(t)$ and $\vec{N}(t)$, and is also a unit vector.

Definition 7.2.6 (Normal and osculating planes)

The **normal plane** is the plane determined by $\vec{N}$ and $\vec{B}$, and contains all lines orthogonal to $\vec{T}$.

The **osculating plane** is the plane determined by $\vec{T}$ and $\vec{N}$, and is the plane closest to containing the part of the curve near $\vec{P} := \vec{r}(t)$.

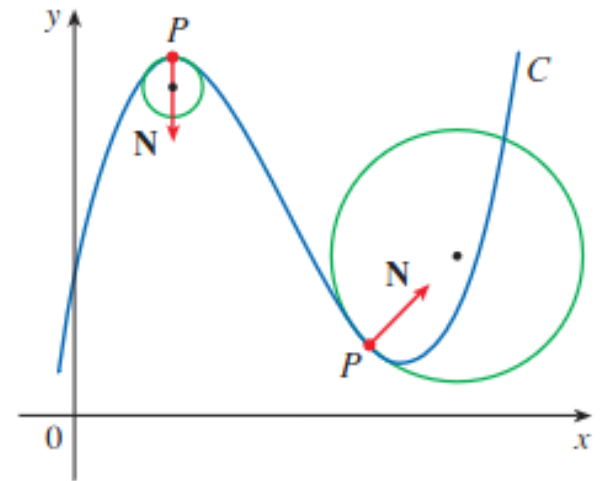


Definition 7.2.7 (Osculating circle)

The **circle of curvature** or **osculating circle** of $C := \vec{r}(I)$ at $\vec{P} := \vec{r}(t)$ is the circle in the osculating plane that passes through $\vec{P}$ with radius $\frac{1}{\kappa}$ and center a distance $\frac{1}{\kappa}$ from $\vec{P}$ along the vector $\vec{N}$.

The center of said circle is called the **center of curvature**.

The circle of curvature shares the same tangent, normal, and curvature with the curve C at P .



Definition 7.2.8 (Torsion)

The **torsion** of a curve is

$$\tau(t) := -\frac{d\vec{B}}{ds} \cdot \vec{N} = -\frac{\vec{B}'(t) \cdot \vec{N}(t)}{|\vec{r}'(t)|} = \frac{(\vec{r}'(t) \times \vec{r}''(t)) \cdot \vec{r}'''(t)}{|\vec{r}'(t) \times \vec{r}''(t)|^2}.$$

The torsion measures how much the curve “twists”; if τ is positive, the curve twists out of the osculating plane at P in the direction of the binormal, and if it’s negative, it twists in the direction opposite the binormal.

7.3. Kinematics

Definition 7.3.1

Let $\vec{r}(t)$ be the position of a particle at time t .

Its **velocity** at time t is $\vec{v}(t) := \vec{r}'(t)$. Its **acceleration** at time t is $\vec{a}(t) := \vec{r}''(t)$.

Its **speed** at time t is $v(t) := |\vec{r}'(t)|$.

Theorem 7.3.2 (Tangential and normal components of acceleration)

$$\vec{v} = v\vec{T}.$$

$$\vec{a} = v'\vec{T} + \kappa v^2 \vec{N}.$$

8. Vector fields + line integrals

See NMD [17-20], Stewart [16.1-16.3]

8.1. Vector fields

Definition 8.1.1 (Vector field)

A vector field in $\mathbb{R}^n$ is a mapping $F : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$.

Definition 8.1.2 (Flow line)

A flow line of a vector field $F : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a differentiable path $\vec{x} : I \rightarrow \mathbb{R}^n$ (where I is an interval on $\mathbb{R}$) such that

$$\vec{x}'(t) = F(\vec{x}(t))$$

That is, the velocity vector of $\vec{x}$ at time t is given by the value of the vector field F at the point on x at time t .

Procedure 8.1.3 (Approximating flow line)

Begin at a vector along the flow line. Then the next vector on the vector field in the direction pointed to by this vector is approximately along the flow line. So you can draw a curve through the vectors following the arrows.

Definition 8.1.4 (Conservative vector field)

A vector field $\vec{F}$ is conservative iff there exists $f : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $\vec{F} = \nabla f$ at all points in $\mathbb{R}^n$. Then f is called the **potential function** for F .

Theorem 8.1.5 (Converse of Poincaré's lemma)

If $\vec{F} : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a continuously differentiable (C^1) conservative vector field, then

$$\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i} \quad \text{for all } i \neq j.$$

Equivalently, throughout D , $J\vec{F} = (J\vec{F})^T$ (the Jacobian is symmetric).

Theorem 8.1.6 (Poincaré's lemma)

If $\vec{F} : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a continuously differentiable (C^1) vector field. If D is simply connected and

$$\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i} \quad \text{for all } i \neq j,$$

or equivalently $J\vec{F} = (J\vec{F})^T$ (the Jacobian is symmetric), then $\vec{F}$ is conservative.

Note: In $\mathbb{R}^3$, $J\vec{F} = (J\vec{F})^T$ is equivalent to $\vec{F}$ being irrotational ($\nabla \times \vec{F} = 0$).

8.2. Line integrals

See https://youtu.be/WA5_a3C2iqY?si=a99FLFGMVS5xyEUM

Definition 8.2.1 (Scalar line integral)

If C is a smooth curve defined by $\vec{r} = \vec{r}(t)$, $a \leq t \leq b$, where $\vec{r} \in \mathbb{R}^n$, then

$$\int_C f(\vec{r}) \, ds = \int_a^b f(\vec{r}(t)) |\vec{r}'(t)| \, dt.$$

Definition 8.2.2 (Line integral of a vector field along a smooth curve)

If F is any continuous vector field defined on a smooth curve C defined by $\vec{r}(t)$, $a \leq t \leq b$, then

$$\int_C \vec{F} \cdot d\vec{r} = \int_C \vec{F} \cdot \hat{T} \, ds = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) \, dt$$

It represents the work done by moving a particle along the curve C , if F is a force field.

Definition 8.2.3 (Path independence)

A vector field $\vec{F}$ is path-independent iff for any two paths C_1 and C_2 which have the same start and end points,

$$\int_{C_1} \vec{F} \cdot d\vec{r} = \int_{C_2} \vec{F} \cdot d\vec{r}$$

Theorem 8.2.4 (Fundamental Theorem of Line Integrals)

Let C be a smooth curve given by the function $\vec{r}(t)$ defined on the interval $t \in [a, b]$. Let f be a differentiable function whose gradient ∇f is continuous on C . Then

$$\int_C \nabla f \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$$

In other words, the line integral of a conservative field along a curve is equal to the potential difference between its endpoints.

This implies that the following statements are equivalent:

- The vector field $\vec{F}$ is conservative
- The vector field $\vec{F}$ is path-independent
- $\oint_C \vec{F} \cdot d\vec{r} = 0$ for every closed path C .

Example. The electric field caused by a single point charge q is

$$\vec{E}(x) = \frac{\varepsilon_0 q}{|\vec{x}|^3} \vec{x}.$$

It's a conservative field, since $-V$ is its potential function:

$$\vec{E} = -\nabla V.$$

$\int_C \vec{E} \cdot d\vec{s}$ is the work per unit charge moving along the path C , and is equal to the voltage difference (by the Fundamental Theorem of Line Integrals):

$$\int_C \vec{E} \cdot d\vec{s} = V(\vec{B}) - V(\vec{A}),$$

where $\vec{A}$ and $\vec{B}$ are the endpoints of C .

9. Multiple integrals

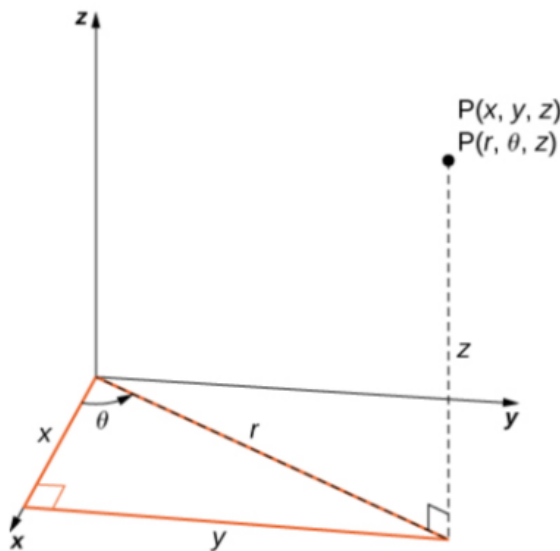
9.1. Coordinate system

Definition 9.1.1 (Polar coordinates)

$$\begin{aligned} x &= r \cos \theta & y &= r \sin \theta \\ r^2 &= x^2 + y^2 & \tan \theta &= \frac{y}{x} \end{aligned}$$

Definition 9.1.2 (Cylindrical coordinates)

$$\begin{aligned} x &= r \cos \theta & y &= r \sin \theta & z &= z \\ r^2 &= x^2 + y^2 & \tan \theta &= \frac{y}{x} & z &= z \end{aligned}$$



Definition 9.1.3 (Spherical coordinates)

Let $\vec{x}$ be a point, and call the line from $\vec{0}$ to $\vec{x}$ the “radius”. It has three coordinates:

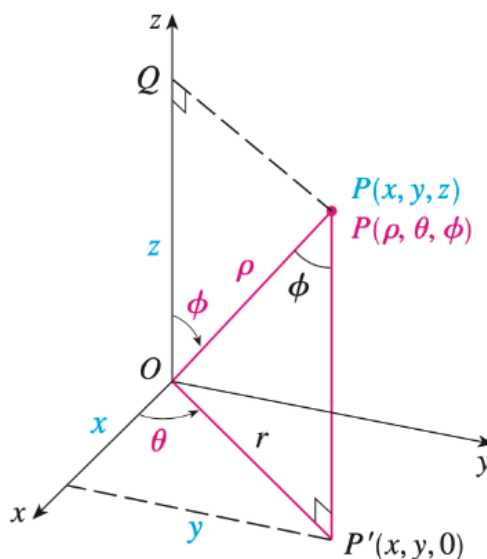
- ρ (or r), the **radial distance** — length of the radius
- θ , the **azimuthal angle** of the radius around the polar axis z , within the xy -plane
- φ , the **polar angle** or **inclination angle** between the radius and the polar axis z

Physicists use φ for the azimuthal angle and θ for the polar angle.

$$x = \rho \sin \varphi \cos \theta \quad y = \rho \sin \varphi \sin \theta \quad z = \rho \cos \varphi$$

$$\rho^2 = x^2 + y^2 + z^2 \quad \varphi = \cos^{-1} \left(\frac{z}{\rho} \right)$$

$$\theta = \cos^{-1} \left(\frac{x}{\rho \sin \varphi} \right)$$



The following may be useful:

°	rad	sin	cos	tan
0°	0	0	1	0
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	1	0	undef

$$\cos^2 x + \sin^2 x = 1.$$

9.2. Double integral

See Colley [5.1, 5.2], Paul’s Notes [15.1, 15.2, 15.3], Brummet [08, MVCWUP:Feb3(29-33)]

Theorem 9.2.1 (Fubini’s Theorem)

If $f : X \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ is continuous on $[a, b] \times [c, d]$, then

$$\begin{aligned} \iint_R f(x, y) \, dA &= \int_a^b \int_c^d f(x, y) \, dy \, dx \\ &= \int_c^d \int_a^b f(x, y) \, dx \, dy \end{aligned}$$

Theorem 9.2.2 (Reversing the order of integration)

Sketch the bounds of the region of integration, then just redo the bounds from scratch.

Theorem 9.2.3 (Converting double integrals to polar)

$$\begin{aligned} \int_a^b \int_c^d f(x, y) \, dx \, dy &= \int_\alpha^\beta \int_\gamma^\delta f(x, y) \, r \, dr \, d\theta \\ &= \int_\alpha^\beta \int_\gamma^\delta f(r \cos \theta, r \sin \theta) \, r \, dr \, d\theta \end{aligned}$$

$a, b, c, d, \alpha, \beta, \gamma, \delta$ may be constants or may depend on the variables. To find the new bounds sketch the bounds and redo the bounds.

Warning: Don't forget the r in $\iint f r dr d\theta$!

9.3. Triple integral

See Stewart [15.6-15.8], Trimm [5.6, 5.7],

Definition 9.3.1 (Triple integral)

The triple integral of f over the region R is

$$\iiint_R f dA = \lim_{n \rightarrow \infty} \sum_{(x_i^*, y_j^*, z_k^*) \in R} f(x_i^*, y_j^*, z_k^*) \Delta A$$

(choose n points (x_i^*, y_j^*, z_k^*) in R , then sum each $f(x_i^*, y_j^*, z_k^*) \Delta A$)

Theorem 9.3.2 (Fubini's Theorem)

If $f : X \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}$ is continuous on $[a, b] \times [c, d] \times [r, s]$, then

$$\iiint_R f(x, y) dA = \int_r^s \int_c^d \int_a^b f(x, y) dx dy dz$$

We have type I, II, and III regions in 3-space just like in the plane, and we integrate just like with double integrals.

Theorem 9.3.3 (Triple integrals in cylindrical coordinates)

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be continuous and

$$D := \{(r, \theta) : \alpha \leq \theta \leq \beta, \quad h_1(\theta) \leq r \leq h_2(\theta)\}$$

$$E := \{(x, y, z) : (x, y) \in D, \quad u_1(x, y) \leq z \leq u_2(x, y)\}.$$

Then,

$$\iiint_E f(x, y, z) dV =$$

$$\int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r \cos \theta, r \sin \theta)}^{u_2(r \cos \theta, r \sin \theta)} f(r \cos \theta, r \sin \theta, z) r dz dr d\theta$$

In short,

$$\iiint f dV = \iiint f r dz dr d\theta.$$

Theorem 9.3.4 (Triple integrals in spherical coordinates)

In short,

$$\iiint f dV = \iiint f \rho^2 \sin \varphi d\rho d\theta d\varphi$$

9.3.1. General change of variables

Theorem 9.3.5 (General change of variables)

Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ which takes in an argument in the coordinate system X , and $T_{XU} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a transformation from coordinates in X to coordinates in U ,

$$\begin{aligned} & \int \cdots \int_R f(\vec{x}) dx_1 \cdots dx_n \\ &= \int \cdots \int_R f(T_{XU}(\vec{u})) |\det JT_{XU}^{-1}| du_1 \cdots du_n \end{aligned}$$

where $|\det JT_{XU}^{-1}|$ is the absolute determinant of the Jacobian of T_{XU}^{-1} (the transformation from coordinates in U to coordinates in X), which is also denoted by

$$|\det JT_{XU}^{-1}| = |\det JT_{UX}| = \left| \frac{\partial(x_1, \dots, x_n)}{\partial(u_1, \dots, u_n)} \right|$$

It can also be computed more easily by taking the inverse of JT_{XU} :

$$|\det JT_{XU}^{-1}| = \left| \frac{1}{\det JT_{XU}} \right|$$

Example: Cartesian to spherical. Converting from Cartesian coordinates to spherical coordinates in $\mathbb{R}^3$:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = T_{XP}^{-1} \left(\begin{bmatrix} \rho \\ \theta \\ \varphi \end{bmatrix} \right) = \begin{bmatrix} \rho \sin \varphi \cos \theta \\ \rho \sin \varphi \sin \theta \\ \rho \cos \varphi \end{bmatrix}$$

so then the absolute Jacobian determinant is

$$|\det JT_{XP}^{-1}| = \rho^2 \sin \varphi$$

10. Div, curl, Green's theorem

10.1. Divergence and curl

See 3Blue1Brown video

Definition 10.1.1 (Del operator)

In $\mathbb{R}^3$, del is defined by

$$\nabla := \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix}$$

In $\mathbb{R}^n$, del is defined by

$$\nabla := \begin{bmatrix} \frac{\partial}{\partial x_1} \\ \vdots \\ \frac{\partial}{\partial x_n} \end{bmatrix}$$

Del is an operator; it takes in a function and outputs a function.

Definition 10.1.2 (Divergence)

Let $\vec{F} : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a differentiable vector field. Then the divergence of $\vec{F}$ is the scalar field

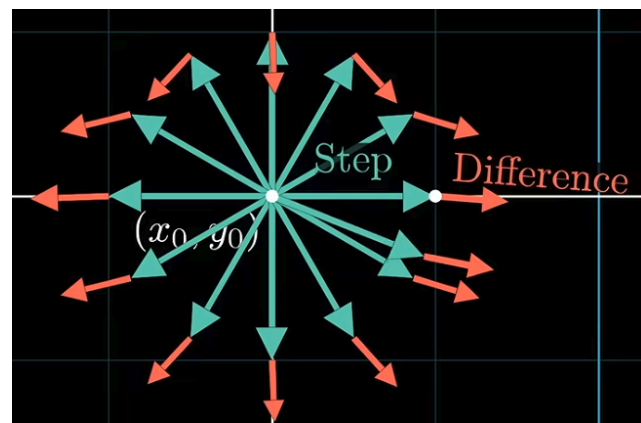
$$\operatorname{div} \vec{F} = \nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x_1} + \cdots + \frac{\partial F_n}{\partial x_n}$$

Lemma 10.1.3

If $\vec{F}$ represents the flow rate of a fluid, then $\operatorname{div} \vec{F}$ represents the net mass flow through each point in the domain of $\vec{F}$:

- If $\operatorname{div} \vec{F} > 0$, then more fluid is flowing out than in.
- If $\operatorname{div} \vec{F} < 0$, then more fluid is flowing in than out.
- If $\operatorname{div} \vec{F} = 0$, then the same amount of fluid flows in as flows out. In this case, $\vec{F}$ is considered **incompressible** and **solenoidal**.

The divergence of a vector field represents how “outgoing” the field is at each point, and how source-like (if positive) or sink-like (if negative) each point is.



The divergence is the average of $\text{Step} \cdot \text{Difference}$ over all directions, with an infinitesimally small step.

Definition 10.1.4 (Curl ($\mathbb{R}^3$))

Let $\vec{F} : X \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a differentiable vector field on $\mathbb{R}^3$. Then the curl of $\vec{F}$ is the vector field

$$\operatorname{curl} \vec{F} = \nabla \times \vec{F}$$

Definition 10.1.5 (Curl ($\mathbb{R}^2$))

Let $\vec{F} : X \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a differentiable vector field on $\mathbb{R}^2$. Then the curl of $\vec{F}$ is the scalar field

$$\operatorname{curl} \vec{F} = \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}$$

This is equivalent to the magnitude of $\nabla \times \vec{F}$, where counterclockwise is positive and clockwise is negative (by right-hand rule).

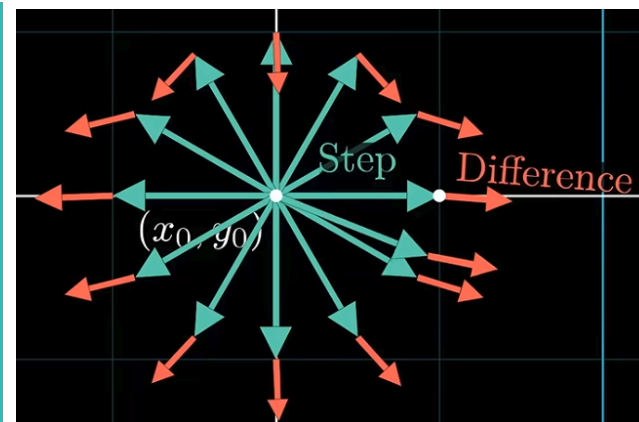
Definition 10.1.6 (Irrotational)

If $\nabla \times \vec{F} = 0$ everywhere on the vector field $\vec{F} : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$, then $\vec{F}$ is considered **irrotational**.

Lemma 10.1.7

Let there exist an infinitesimally small sphere at the point $\vec{x} \in X$. Let $\vec{F} : X \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a vector field that represents the velocity of a fluid at each point in X . Then $\operatorname{curl} \vec{F}$ is the unique vector such that

- The direction of $\operatorname{curl} \vec{F}$ is along the axis of rotation of the sphere, following the right-hand rule.
- The magnitude of $\operatorname{curl} \vec{F}$ is the speed of the rotation of the sphere.



The curl is the average of $\text{Step} \times \text{Difference}$ over all directions, with an infinitesimally small step.

10.2. Green's theorem

See Video

See NMD [30, 31], Stewart [16.4]

Theorem 10.2.1 (Green's Theorem)

Let ∂D be a positively oriented (counterclockwise, i.e. D is to the left as you follow ∂D), piecewise smooth, simple closed curve in the xy -plane, and D be the region bounded by ∂D . If P and Q have continuous partial derivatives on an open region containing D , then

$$\oint_{\partial D} P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Equivalently, if $\vec{F} = x, y \mapsto \begin{bmatrix} P(x,y) \\ Q(x,y) \end{bmatrix}$, then

$$\oint_{\partial D} \vec{F} \cdot d\vec{r} = \iint_D |\nabla \times \vec{F}| dA$$

In other words, the circulation of a vector field along a curve is the same as the sum of the curls within the region bounded by the curve.

Definition 10.2.2 (Circulation)

The circulation of the vector field $\vec{F}$ around the curve C is

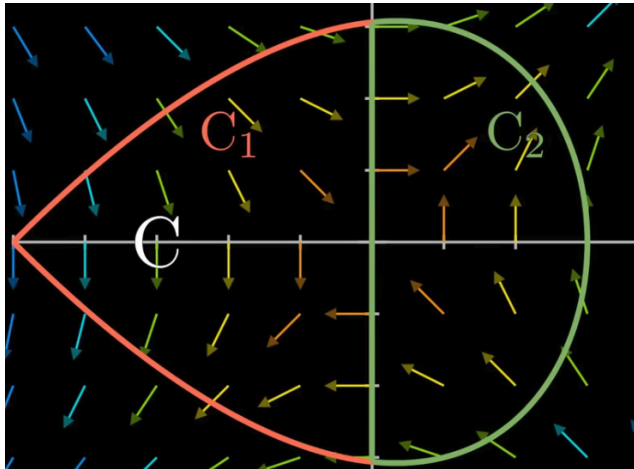
$$\oint_C \vec{F} \cdot d\vec{r}$$

It measures how much F aligns with the curve C .

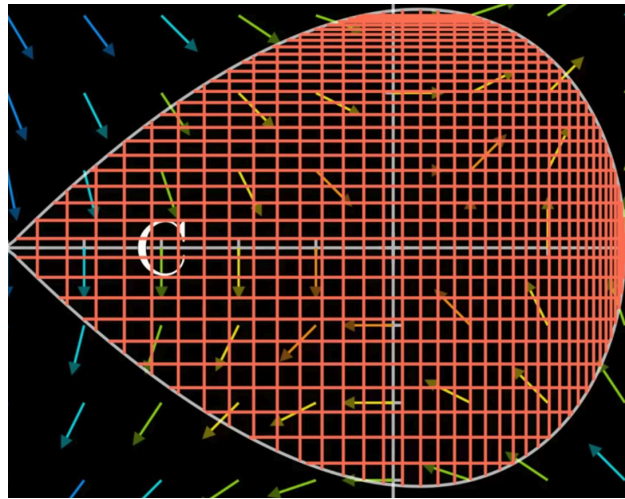
Procedure 10.2.3 (Motivation of Green's Theorem)

Let D be a connected region bounded by $\partial D = C$, and D_1 and D_2 be two disjoint connected halves of that region with ∂D_1 and ∂D_2 . Then

$$\oint_C f \, ds = \oint_{C_1} f \, ds + \oint_{C_2} f \, ds.$$



Now consider the circulation $\oint_C \vec{F} \cdot d\vec{r}$, and partition D into infinitely many infinitesimally small areas dA :



The circulation around an infinitesimally small area is the same as the curl on that area, so

$$\oint_{\partial D} \vec{F} \cdot d\vec{r} = \iint_D |\nabla \times \vec{F}| \, dA.$$

11. Surfaces

11.1. Parametric surfaces

See NMD [28], Stewart [16.6]

Definition 11.1.1 (Parametric/parameterized surface)

Let $\vec{X} : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a one-to-one function (except possibly at the boundary of D). Then the image of X is called a parameterized surface.

Definition 11.1.2 (Normal vector to a parameterized surface)

Let $\vec{X} = \begin{bmatrix} u \\ v \end{bmatrix} \mapsto \begin{bmatrix} x(u,v) \\ y(u,v) \\ z(u,v) \end{bmatrix}$ be a parameterization of a surface, and let $\vec{u}$ be a vector in the domain of $\vec{X}$. Then the tangent vector along the u -axis is $\vec{X}_u(\vec{u})$, where

$$\vec{X}_u = \begin{bmatrix} \frac{\partial x}{\partial u} \\ \frac{\partial y}{\partial u} \\ \frac{\partial z}{\partial u} \end{bmatrix}$$

and similarly the tangent vector along the v -axis is $\vec{X}_v(\vec{u})$.

Then the normal vector to the parameterized surface at the point $\vec{u}$ is

$$\vec{N} = \vec{X}_u(\vec{u}) \times \vec{X}_v(\vec{u})$$

Definition 11.1.3 (Smooth)

A parameterization $\vec{X}$ of a surface is smooth at a point if its normal vector is not equal to 0 at that point.

A surface is smooth at a point if there exists a parameterization for that surface which is smooth at that point.

Note that a smooth surface can have non-smooth parameterizations.

11.2. Surface integrals

See NMD [29], Stewart [16.7]

Definition 11.2.1 (Scalar surface integral)

The surface integral of f over the surface S which is parameterized by $\vec{X} : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3 = (u, v) \rightarrow (x, y, z)$ and where D is the domain of $\vec{X}$ is

$$\iint_S f \, dS = \iint_D f(\vec{X}(u, v)) |\vec{X}_u \times \vec{X}_v| \, dA$$

(dS is a part of the surface area, dA is a part of the domain)

Definition 11.2.2 (Scalar surface integral in $\mathbb{R}^3$ for function of two variables)

(This is optional; the definition above can be used to derive this.)

The surface integral of f over the surface S defined by $z = g(x, y)$, whose domain is D , is

$$\iint_S f \, dS = \iint_D f(x, y, g(x, y)) \sqrt{\left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2 + 1} \, dA$$

Definition 11.2.3 (Orientable surface)

A smooth, connected surface S is orientable iff it is possible to define a single normal vector at each point of S such that the collection of these normal vectors varies continuously over S .

Definition 11.2.4 (Closed surface)

A surface is closed iff it is the boundary of some solid region E .

Definition 11.2.5 (Oriented surface)

A smooth, orientable surface together with a choice of its orientation.

If the surface is closed and encloses the region E , then it has a **positive orientation** when we choose the set of its normal vectors to point away from E and a **negative orientation** when we choose the set of its normal vectors to point towards E .

Definition 11.2.6 (Vector surface integral, flux)

The surface integral or flux of the vector field $\vec{F}$ over the surface S which is parameterized by $\vec{X} : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3 = (u, v) \rightarrow (x, y, z)$ and where D is the domain of $\vec{X}$ is

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_D F(\vec{X}(u, v)) \cdot (\vec{X}_u \times \vec{X}_v) \, du \, dv$$

11.3. Stokes' Theorem

Definition 11.3.1 (Positive oriented boundary)

Given a surface S whose boundary is the curve C , the positive orientation of the curve is such that if one were to walk along the curve in that direction, with the vector upwards from the top of their head parallel to the normal vectors of S , the surface would be to that person's left. Equivalently, you could choose a normal vector and use the right-hand rule (thumb is the normal vector and the positive orientation is given by the curl of the fingers).

Theorem 11.3.2 (Stokes' Theorem)

Let S be an oriented smooth surface, bounded by a curve ∂S , composed of finitely many simple closed smooth differentiable (C^1) curves with positive orientation. Let $\vec{F}$ be a differentiable (C^1) vector field whose domain includes S . Then

$$\oint_{\partial S} \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

In other words, the circulation of a vector field along a curve is the same as the sum of the curls within the surface bounded by the curve.

11.4. Divergence theorem

Theorem 11.4.1 (Gauss's Theorem / Divergence theorem)

Let D be a solid region in $\mathbb{R}^3$, bounded by a surface ∂D , composed of finitely many smooth closed surfaces with positive orientation. Let $\vec{F}$ be a differentiable (C^1) vector field whose domain includes D . Then

$$\oiint_{\partial D} \vec{F} \cdot d\vec{S} = \iiint_D \nabla \cdot \vec{F} \, dV$$

In other words, the flux of a vector field through a closed surface is the same as the sum of the divergences of the vector field through the region bounded by the surface.

This follows from that

- The ratio of flux to volume approaches the divergence as the volume becomes smaller.
- If the region is partitioned into smaller regions, the flux of the region is equal to the sum of the flux of the smaller regions (since the flux of the boundary of the two regions cancels out).